Hierarchical Prototype Learning With Uncertainty-Aware Adaptation for Cross-Domain Semantic Segmentation of Remote Sensing Images

Jiawei Ning, Zhongle Ren, *Member, IEEE*, Biao Hou, *Senior Member, IEEE*, Runnong Jiang, Shijie Liu, Weibin Li, and Licheng Jiao, *Life Fellow, IEEE*

***Abstract*—Prominent domain discrepancies in remote sensing images (RSIs), such as sensor types, geographical patterns, and land usage, significantly hinder the research and practical applications of cross-scene land classification. Unsupervised domain adaptation (UDA) fully exploits domain invariance between the labeled source and unlabeled target domain (TD), which alleviates the challenge of inaccurate land classification due to a lack of labels in RSIs. However, most existing UDA methods for RSI semantic segmentation are insufficient in exploring cross-domain features and have difficulty in modeling fine-grained domain-invariant features between interclass. In this article, we propose a novel self-training UDA method named hierarchical prototype learning (HPL), which learns the inherent domain-wise consistency and classwise invariance through progressive exploration of the prototypy, significantly alleviating the ambiguity in uncertain regions. HPL mainly consists of domainwise progressive prototype interaction (DPPI) and classwise dynamic prototype collaboration (CDPC). DPPI and CDPC specialize in hierarchically building prototype interaction architectures tailored to domainwise alignment and classwise calibration, respectively. This design not only mitigates the sensitivity to cross-domain scenarios but also allows for the precise correction of uncertain regions. Furthermore, CDPC exhibits the capacity for pixel-level category restoration and promotes the correct and fine-grained updating of pseudolabels. Extensive experiments on two public datasets and a private self-built dataset demonstrate the superiority of HPL over other state-of-the-art (SOTA) methods for UDA semantic segmentation of RSIs.**

***Index Terms*—Class-level prototype guidance, domain-level prototype alignment, multilevel prototype learning, unsupervised domain adaptation (UDA).**

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Jiawei Ning, Zhongle Ren, Biao Hou, Runnong Jiang, Shijie Liu, and Licheng Jiao are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, and the Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: jwning@stu.xidian.edu.cn; zlren@xidian.edu.cn; avcodec@hotmail.com; rn_jiang@163.com; sjie_liu@163.com; lchjiao@mail.xidian.edu.cn).

Weibin Li is with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, and the Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China, and also with Hangzhou Institute of Technology, Xidian University, Hangzhou 311200, China (e-mail: weibinli@xidian.edu.cn).

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I. Introduction

REMOTE sensing images semantic segmentation (RSISS) is a vital task for land use and land-cover mapping, which aims to finely implement pixel-level semantic category assignment of remote sensing images (RSIs) [1]. It is crucial for the effective management of urban spatial planning, ecological environment monitoring, natural resource auditing, and disaster emergency response [2], [3]. Recently, deep-learning-based semantic segmentation methods have provided advanced solutions for efficient interpretation. However, there are two main problems in remote sensing data hindering the enormous potential of intelligent interpretation for RSISS.

1) With the continuous enrichment of the quantity and quality of high-resolution remote sensing imagery, the domain discrepancies between various scenes in remote sensing imagery are significantly magnified (such as different sensors, land cover, and land form as shown in Fig. 1).
2) Annotating remote sensing imagery extensively is a time-consuming and labor-intensive task, which makes it impractical to extract the global land use and land-cover annotation by manual procedure.

These issues lead to the struggling segmentation effectiveness and evident constraint in real-world scenarios of fully-supervised models under unlabeled data [4]. Therefore, enhancing the model's generalization competence with limited labeled data has become a crucial research focus.

In recent years, most of the domain-specific pretrained foundation models [5], [6], [7] outperformed the general foundation models, which indicates that pretraining within specific tasks (e.g., remote sensing and medical image) is particularly crucial for uncovering the latent feature extraction capability of the backbone network. Nevertheless, the pretraining on large-scale and extensive data is effective at enhancing the coarse capabilities (e.g., classification, segmentation, and detection) of the model during the fine-tuning stage, instead of exploration in specific downstream tasks (e.g., semantic domain adaptation and generalization).

Unsupervised domain adaptation (UDA) leverages the limited labeled data from the source domain (SD) to learn invariant features between the SD and the unlabeled target domain (TD), with the aim of effectively mitigating domain discrepancies. It ensures that reliable semantic segmentation

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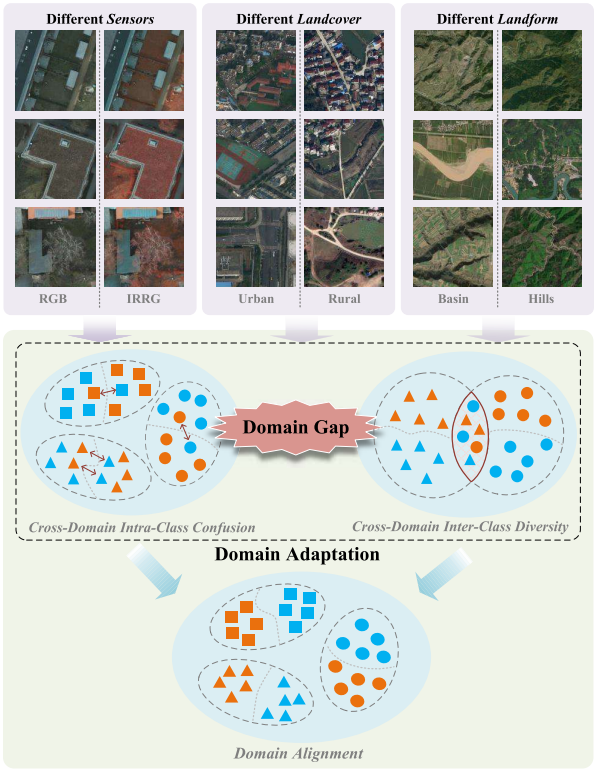


Fig. 1. Illustration of the three different domain discrepancies, which are different sensors, land cover, and land form. Domain-adaptive correlation methods are widely used to overcome the domain gap produced by the reasons mentioned above, which include both intraclass confusion and interclass diversity in cross-domain scenarios. Different shapes represent various classes, and different colors denote categories from different domains.

performance is maintained in the unlabeled TD. To conduct precise research on domain discrepancy, UDA methods for RSIs have been extensively studied. For image-level classification tasks, methods such as distance metric learning or adversarial learning frameworks provide cross-domain feature alignment. The pixelwise classification inherent to RSISS, a dense prediction task, renders the modeling of invariant features particularly challenging. Hence, minimizing domain shift demands style-invariant mapping across domains coupled with class-conditional feature consistency enforcement [8]. Currently, the predominant deep-learning-based UDA approaches for RSISS are mainly divided into two types: adversarial learning and self-training.

In terms of adversarial learning, a discriminator is typically constructed to obscure the distinctive features between the source and TD, thereby equipping the model with the capability to generalize features on unseen data. Inspired by UDA tasks on natural images [9], [10], [11], various adversarial strategies and reliable network architectures are introduced to reduce interdomain discrepancies [12], [13], [14], [15]. These methods exhibit a global approach in mitigating scene-level domain discrepancies, particularly suitable for image classification [16], [17], [18], [19], [20], and object detection [21], [22]. However, in complex RSISS scenarios, conventional unilateral adversarial training fails to enforce intraclass feature alignment, thereby inducing a pronounced generalization bottleneck.

Regarding self-training, it generates a pseudolabel for the TD, which allows for pixel-level self-supervised learning, facilitating the in-depth exploration of cross-domain features [23], [24], [25]. The majority of contemporary UDA methods either extend adversarial learning frameworks or establish standalone self-training systems to generate such labels. In RSISS, [26], [27], [28], [29] achieved notable progress through diverse mechanisms for pseudolabel refinement. It is evident that the quality of pseudolabel is a determining factor in the model's ability to generalize in the TD [30]. Existing research predominantly generates pseudolabels through threshold control mechanisms, yet this methodology suffers from notable drawbacks. A primary issue is that most pseudolabel filtering methods rely on static thresholds, which hinder the improvement of pseudolabel quality in intricate scenes. Furthermore, the correlation between-class posterior probabilities and the fundamental cross-domain dependencies remains underexplored, leading to difficulties in updating low-confidence regions within pseudolabels.

In summary, existing UDA methods of RSISS include the following problems.

1) Existing methods inadequately address the architectural discrepancies of backbone networks across remote sensing domains, while the representational misalignment between hierarchical modules remains unresolved.
2) The reliance of self-training on fixed thresholds for pseudolabel optimization inherently limits its capacity to adapt to dynamic cross-domain feature distributions, thereby impairing generalization performance in unseen scenarios.

To address the aforementioned issues, considering the representative features of the prototypes, we propose a novel self-training UDA method for RSISS in this article, named hierarchical prototype learning (HPL), which systematically resolves the hierarchical effectiveness of prototypes in mitigating cross-domain discrepancies. The proposed HPL consists of domainwise progressive prototype interaction (DPPI) and classwise dynamic prototype collaboration (CDPC). In particular, the DPPI methodically investigates cross-layer feature transferability in remote sensing-specific domains through shallow-to-deep prototype interaction, which coordinates domain-invariant representations across multilayer features. Furthermore, CDPC is specifically designed to optimize pseudolabels in self-training. Dynamic prototype extraction integrates prototypical knowledge of classwise features from the source and the TD to reduce intraclass disparity. Adaptive uncertainty-aware calibration achieves precise pseudolabel refinement in uncertain pixels by a nonfixed threshold and class-specific prototype prior. Two public datasets and a private dataset were employed in comprehensive experimental investigations. The results therefrom unequivocally demonstrate the preeminence of HPL over other state-of-the-art (SOTA) approaches in the context of UDA RSISS.

In summary, the main contributions of the proposed method can be summarized as follows.

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1) In this article, a novel hierarchical prototype alignment framework, HPL is proposed. The framework incorporates the DPPI and CDPC, which are designed to progressively mitigate cross-domain disparities.
2) DPPI completely alleviates the domain shift in the feature extraction from the perspective of remote sensing through shallow-to-deep prototype interaction and domainwise alignment.
3) Benefiting from the dynamic classwise prototype prior and the adaptive uncertainty-aware calibration, CDPC accomplishes meticulous pixel-level pseudolabel refinement.

The rest of this article is organized as follows. Section II makes a description of the existing UDA semantic segmentation works. Section III elaborates on the details of the proposed HPL. Experimental setup and discussion about the experimental results have been demonstrated in Section IV. Finally, Section V concludes this article and proposes future work.

II. Related Work

In this section, the works related to the proposed HPL are presented, which include UDA, UDA in RSISS, and prototype learning in RSISS.

A. Unsupervised Domain Adaptation

To address the problem of limited generalizability of fully-supervised methods in cross-domain, a series of work [31], [32], [33] by Wang et al. made a prominent contribution to the interpretability of interdomain differences by calculating and optimizing the absolute distance between source and TD from the perspective of statistical information. Following this, most of the work [34], [35], [36], [37] used adversarial networks to learn global domain variability, modeling domain-invariant features to guide the stability of the network's performance in the target. However, in semantic segmentation tasks, it is difficult to model pixel-level category features locally using adversarial learning alone. Considering the TD in which self-supervised learning can more completely utilize the predictive power of deep neural networks, approaches using self-training strategies have been widely developed [38], [39], [40], with a large number of studies exploring the problem around how to generate better-quality pseudolabels. Hoyer et al. [38] proposed various strategies for data augmentation, including rare class sampling, contextual feature incorporation [39], and mask learning [40], and various strategies for self-training, all of which achieved excellent generalization performance in the TD. Besides, [41], [42], [43], [44], [45], [46] also constructed effective self-training frameworks through different perspectives. Despite these methods demonstrating superiority in the natural image domain, they lack competitiveness in cross-domain remote sensing imagery.

B. UDA in RSISS

Unlike natural images, the cross-domain problems in RSISS exhibit characteristics of scale variability, scene diversity, and distinct styles [47]. Adversarial domain adaptation plays a positive role in reducing domain discrepancies. However, it is not capable of modeling cross-domain features with fine-grained detail at the class level. Benefiting from the effectiveness of self-training, extant methods [26], [27], [28], [29] address the problem of RSISS under UDA in abundant ways. Zhang et al. [26] designed an adaptive hierarchical pseudolabel weight setting method with a self-training paradigm to reduce the effect of label noise while improving the quality of the generated pseudolabels. Xi et al. [27] proposed a pseudolabel generation strategy to effectively combine the strengths of different experts, which can be flexibly applied to two cases where the source class union is equal to or includes the target class set. Gao et al. [28] designed two masked consistency learning strategies to exquisitely explore the contextual structure of target RSIs and improve the quality of pseudolabel for prototype updating. Self-training strategies are limited by their exclusive focus on pixel-level discrete relationships when establishing interclass domain-invariant feature mappings [48]. Although existing methods conduct research in dynamic pseudolabeling generation [49], [50], they are either prohibitively costly in terms of processing [49] or not entirely suitable for cross-domain scenes [50]. Liu et al. [51] proposes an uncertain example mining network with multilevel pseudolabel correcting and uncertainty-based valuable example mining, but its capacity to handle cross-period scenarios and significantly different sensor types, as well as inconsistent cross-domain class spaces. Furthermore, intraclass feature diversity and interclass consistency are insufficiently considered, which impedes the optimal utilization of UDA methods.

C. Prototype Learning in RSISS

Class prototypes, as abstract embeddings of fundamental semantic representations, serve to enhance intraclass compactness and interclass separability [52]. Benefiting from the aforementioned advantages, the prototype-based methods have been effectively applied to the domain of few-shot learning, yielding substantial performance improvements [22], [53], [54]. Zhou et al. [55] introduced a few-shot segmentation network HPA via training-free priors, prototype activation, and cross-referenced decoding to address overfitting and boundary inaccuracies. In recent years, prototype learning has been widely applied in image semantic segmentation. For example, Yu et al. [56] proposed a plug-and-play MGLP that improves performance via multigrained prototypes with hierarchical abstraction and explicit structuralization through horizontal metrics and vertical semantic constraints. In the field of remote sensing, Zhang et al. [26] relied on representative prototypes, the feature centroids of classes, and exploited the feature distances from prototypes, to address the issue for UDA. Jiang et al. [57] incorporated interclass information into class-level prototypes, and adopted the class-centered distribution alignment for adaptation by contrastive learning. Peng et al. [58] proposed a prototype-guided pseudolabel generating approach that generated target pseudolabels based on diverse source prototypes, thereby alleviating the issue of classifier predictions. Furthermore, in the UDA semantic segmentation

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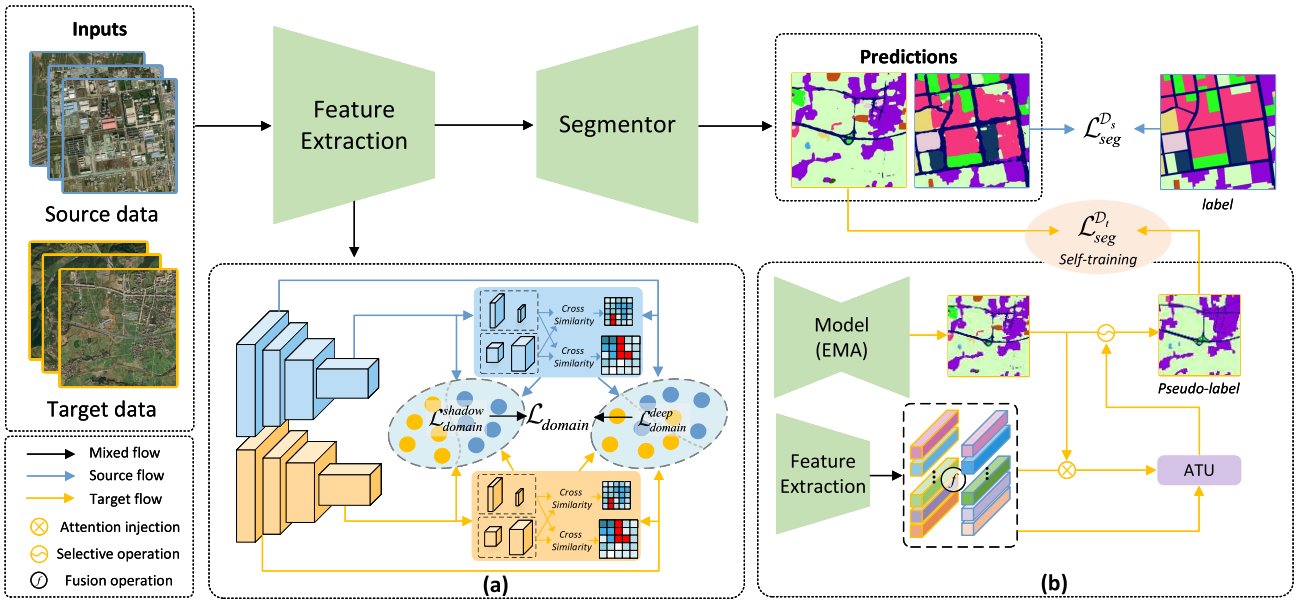


Fig. 2. Self-training framework of the proposed HPL. It consists of two prototype learning pipelines, which are DPPI and CDPC. (a) DPPI incorporates the shallow-to-deep prototype interaction module and the domainwise prototype alignment module. (b) CDPC includes the dynamic classwise prototype extraction module and the adaptive uncertainty-aware calibration module.

of RSIs, Zhu et al. [15] embedded an invariant feature memory module to store invariant domain-level prototype information by a category attention-driven invariant domain-level memory aggregation module, representing the current pseudoinvariant feature. Gao et al. [29] abstracted the cross-domain and intradomain class-level relations into multiple prototypes that the multiple prototypes within the same class can better model rich intraclass relations. Based on this, the cross-domain invariance of domain-level and class-level prototypes is fully explored and effectively extended to the RSISS tasks of UDA in this article.

III. Proposed Method

In this section, five main parts of the proposed HPL are presented: Preliminary, Framework, DPPI, CDPC, and Loss function. Detailed descriptions of these five parts are presented in Sections III-A–III-E.

A. Preliminary

Domain adaptation aims to leverage labeled SD $\mathcal{D}_s$ and unlabeled TD $\mathcal{D}_t$ for feature learning, enabling the model to generalize well in the TD and achieve desirable results. Typically, the SD samples and the TD samples are denoted as $\mathcal{X}_s \subset \mathcal{D}_s$ and $\mathcal{X}_t \subset \mathcal{D}_t$, respectively. They are learned through a deep network encoder E which yields intermediate feature spaces X_s and X_t. Subsequently, the decoder D is employed to obtain the final outputs $\hat{Y}_s$ and $\hat{Y}_t$. This process facilitates the extraction and transformation of features from the source and TD, enabling the model to adapt and generalize effectively. The aforementioned process can be formalized as

$$\widehat{Y}_s, \widehat{Y}_t = D\left(E\left(\mathcal{X}_s, \mathcal{X}_t\right)\right). \tag{1}$$

Due to the fact that the TD is unlabeled, the supervised model from the SD can not be perfectly applied to the TD with a different distribution, which results in the emergence of the following issues:

$$P\left(\mathcal{X}_s\right) \neq P\left(\mathcal{X}_t\right) \tag{2}$$

where the formula indicates that the edge distribution $P(\cdot)$ is unequal in the source and TD. In the problem of remote sensing semantic segmentation, in addition to considering overall domain differences in the images, it is essential to take into account intraclass and interclass domain diversity. Therefore, the objective of optimization is to facilitate the learning of feature invariance between domains, thereby enabling the model to continuously refine its adaptive performance in the TD, which can be defined as

$$\arg\min_{\theta} \left(\Phi\left(X_s, X_t\right) | \theta\right) \tag{3}$$

where $\Phi(\cdot)$ is distance function between X_s and X_t and the domain gap is reduced by the learnable parameters θ. Furthermore, in the context of UDA, the utilization of pseudolabels derived from the TD represents a pivotal approach, that addresses the inherent variability between domains in the dense prediction tasks. This strategy aims to mitigate the challenges posed by the distributional shift between the source and TD, enhancing the model's robustness to generalize and perform accurately in the TD

$$\arg\min_{\theta} \left(\left[L\left(\widehat{Y}_s, Y_s\right) + L\left(\widehat{Y}_t, \tilde{Y}_t\right)\right] \middle| \theta\right) \tag{4}$$

where $L(\cdot)$ is the loss function between prediction and Ground Truth (GT), and $\tilde{Y}_t$ is the pseudolabel of the TD samples. Therefore, optimizing $\tilde{Y}_t$ emerges as one of the primary focal points addressed in this article.

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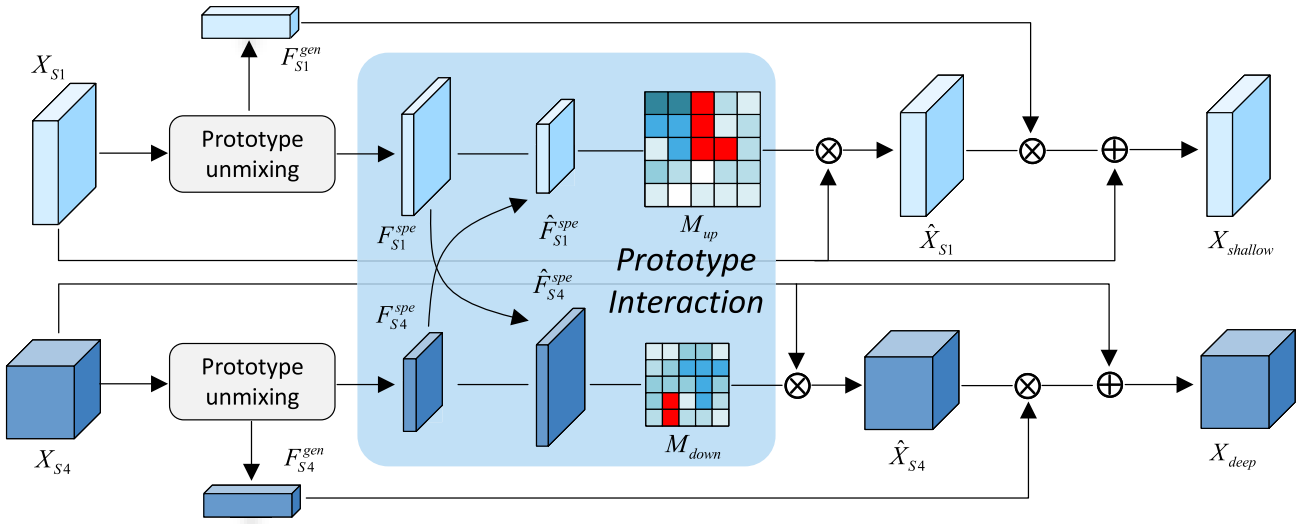


Fig. 3. Framework of the proposed DPPI module.

B. Framework

The overall architecture of the proposed HPL is shown in Fig. 2. It consists of two prototype learning pipelines, which are DPPI and CDPC. In particular, DPPI incorporates the shallow-to-deep prototype interaction (SDPI) module and the domainwise prototype alignment (DPA) module. Moreover, CDPC includes the dynamic classwise prototype extraction (DCPE) module and the adaptive uncertainty-aware calibration (AUC) module. Detailed procedures are described in the following three subsections.

C. Domainwise Progressive Prototype Interaction

We consider that establishing cross-domain feature invariance within the feature extraction contributes to reducing the general domain discrepancy. DPPI is divided into two main parts, which include shallow-to-deep prototype interaction and domainwise prototype alignment as shown in Fig. 3.

1) Shallow-to-Deep Prototype Interaction: In order to thoroughly exploit the local and global cross-domain semantic information for constructing domain-level prototypes, shallow and deep features are utilized to extract prototypes. To alleviate the local-global cross-domain sensitivity of the backbone network, prototype unmixing is utilized to extract the general prototype and specific prototype of the feature. It consists of two operations: average pooling and max pooling. Prototype unmixing operation $\varphi()$ can be defined as

$$\begin{cases} F_{S1}^{\text{gen}}, F_{S1}^{\text{spe}} = \varphi\left(X_{S1}\right) \\ F_{S4}^{\text{gen}}, F_{S4}^{\text{spe}} = \varphi\left(X_{S4}\right) \end{cases} \tag{5}$$

where X_{S1} and X_{S4} denote the shallow feature in the first stage and deep feature in the fourth stage. Hence, the general prototypes (F_{S1}^{gen} and F_{S4}^{gen}) denotes the semantically invariant representation while the specific prototypes (F_{S1}^{spe} and F_{S4}^{spe}) signifies the complex spatial perception.

Then, the cross-block F_{S1}^{spe} and F_{S4}^{spe} maintain dimensional consistency through upsampling and downsampling, and they are employed in prototype interaction via cross-block similarity. It can be a formula as

$$\begin{cases} M_{\text{down}} = \text{Sim}\left(D\left(F_{S1}^{\text{spe}}\right), F_{S4}^{\text{spe}}\right) \\ M_{\text{up}} = \text{Sim}\left(U\left(F_{S4}^{\text{spe}}\right), F_{S1}^{\text{spe}}\right) \end{cases} \tag{6}$$

where $\text{Sim}(\cdot)$, $U(\cdot)$, and $D(\cdot)$ represents the cosine similarity, upscaling and downscaling operation. M_{up} and M_{down} exhibit the correlation of prototypes from the shallow-to-deep features, which integrates the cross-block spatial consistency. Combined with general prototypes, it jointly guides the input feature for domain-invariance learning

$$\begin{cases} X_{\text{shallow}} = F_{S1}^{\text{gen}} \otimes \left(X_{S1} \otimes M_{\text{up}}\right) + X_{S1} \\ X_{\text{deep}} = F_{S4}^{\text{gen}} \otimes \left(X_{S4} \otimes M_{\text{down}}\right) + X_{S4} \end{cases} \tag{7}$$

where X_{shallow} and X_{deep} are innovatively derived through DPPI to positively substitute for the original input X_{S1} and X_{S4}. Furthermore, they are utilized as domainwise features in domain disparity alignment, significantly improving the overall performance.

2) Domainwise Prototype Alignment: The grouped domainwise features of the source and the TD are defined as $(X_{\text{shallow}}^{s}, X_{\text{deep}}^{s})$ and $(X_{\text{shallow}}^{t}, X_{\text{deep}}^{t})$. Therefore, the maximum mean discrepancy [59] distance metric is utilized to measure and align the domainwise discrepancy $\mathcal{L}_{\text{domain}}$

$$\begin{aligned} \mathcal{L}_{\text{domain}} = & \mathcal{L}_{\text{domain}}^{\text{shallow}}\left(X_{\text{shallow}}^{s}, X_{\text{shallow}}^{t}\right) \\ & + \mathcal{L}_{\text{domain}}^{\text{deep}}\left(X_{\text{deep}}^{s}, X_{\text{deep}}^{t}\right). \end{aligned} \tag{8}$$

D. Classwise Dynamic Prototype Collaboration

While the domainwise discrepancy alignment module DPPI effectively mitigates cross-domain shifts, its capability remains limited in adequately capturing fine-grained class distributions and scenario-specific style variations. This limitation highlights the critical need to systematically explore intraclass consistency and interclass discriminability across domains. To address these challenges, we propose a CDPC module, specifically targeted at class-aware domain-invariant

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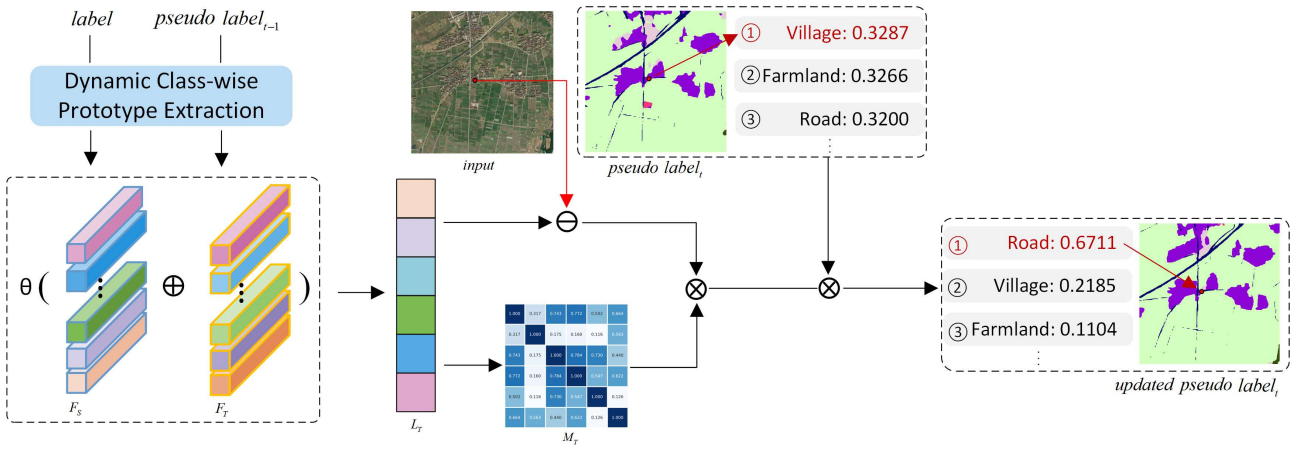


Fig. 4. Framework of the proposed CDPC module. CDPC is proposed to focus on classwise domain-invariant feature alignment and adaptive pseudolabel optimization in self-training, which consists of dynamic classwise prototype extraction and adaptive uncertainty-aware calibration.

feature alignment and uncertainty-guided pseudolabel refinement in self-training, incorporating two coordinated mechanisms: dynamic classwise prototype extraction and adaptive uncertainty-aware calibration, as shown in Fig. 4.

1) Dynamic Classwise Prototype Extraction: In order to obtain the fine-grained classwise prototype representation, RSIs and GTs are used for calculation in the SD, while RSIs and pseudolabels are employed for estimation in the TD. In this way, the domain feature representation of categories can be derived. It is described as

$$P_{\mathcal{D}}^{c} = \frac{\sum_{\mathcal{X}\in\mathcal{D}} \sum_{(i,j)\in\widehat{Y}} \widehat{Y}^{(i,j,c)} \cdot \mathbb{1}\left(Y^{(i,j,c)} == 1\right)}{\sum_{\mathcal{X}\in\mathcal{D}} \sum_{(i,j)\in Y} \mathbb{1}\left(Y^{(i,j,c)} == 1\right)}. \tag{9}$$

Hence, it is feasible to perform intraclass feature fusion of the SD prototypes $P_{\mathcal{D}_s}^{c}$ and TD prototypes $P_{\mathcal{D}_t}^{c}$ to obtain class prototype guidance information P_f^c

$$L_T = \theta(F_S, F_T) = \frac{1}{2}(F_S + F_T) \tag{10}$$

where $\theta(\cdot)$ is denoted as fusion function.

The GT of the SD provides reliable prototype information for each class, which can introduce class-specific features during the fusion process. However, due to domain discrepancy, relying on SD information exclusively increases the risk of model overfitting. In our work, the exponential moving average is utilized to predict the pseudolabels of the TD by prohibiting the backward of gradients

$$\phi_{t+1} \leftarrow \alpha\phi_t + (1-\alpha)\theta_t \tag{11}$$

where α is smoothing factor, which weights both sutudent network f_θ and teacher network g_ϕ. Therefore, pseudolabel Y_{pseudo}^{t+1} can be formula as

$$Y_{\text{pseudo}}^{t+1} = g_\phi\left(X_t|\phi_{t+1}\right). \tag{12}$$

2) Adaptive Uncertainty-Aware Calibration: As depicted in Fig. 5, according to the preset threshold, there are uncertain regions in the prediction results, which typically exhibit two characteristics: manifesting abundant edge structures and interclass features with homogeneous styles. From a probabilistic perspective, uncertain pixels generally present with

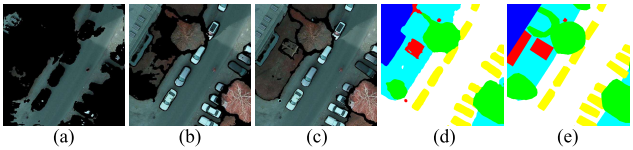


Fig. 5. Illustration of uncertain regions (black color) with adaptive threshold σ on the ISPRS dataset during training. (a) First epoch. (b) 50th epoch. (c) Well-trained. (d) Predict the result. (e) GT.

nearly indistinguishable interclass probability distributions. This manifestation concretely reflects substantial decision boundary degradation, inherently inducing pronounced misclassifications. In the AUC module, prototype distance guidance is strategically leveraged to refine self-adaptive pseudolabels through metric-constrained calibration, enabling the restoration of discriminative class characteristics in feature embeddings.

Given the nonnecessity of updating all pixels for pseudolabel refinement, discriminative identification of optimization-worthy regions becomes imperative

$$\widehat{\mathcal{Y}}_c^i = \left\{\widehat{\mathcal{Y}}_c^i | p_i < \sigma\right\}. \tag{13}$$

In (13), $\mathcal{Y}_c^i$ represents the region where the pseudolabels of the predicted probability p_i are screened out under the threshold σ. However, the majority of existing methods set a fixed threshold or data-specific dynamic threshold for σ, which makes it difficult to achieve appreciable generalization in variable scenarios. Thus, a novel dynamic threshold in this article is designed for the selective process

$$\sigma = b_i * \text{mean}\left(\sum_{c\in N_c} \widehat{\mathcal{Y}}_c^i\right) \tag{14}$$

where b maintains a critical balance, ensuring deterministic calculation of precisely specified update regions across all operational conditions through the smoothing factor ϵ, which is defined as

$$b_i = -\frac{\exp\left(\text{var}\left(\sum_{c\in N_c} \widehat{\mathcal{Y}}_c^i\right)\right)}{\log\left(\text{var}\left(\sum_{c\in N_c} \widehat{\mathcal{Y}}_c^i\right)\right) + \epsilon} \tag{15}$$

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where exp(·), log(·), and var(·) are denoted as exponential, logarithmic, and variance function, respectively.

Then, the feature distance between the TD RSIs and the fused prototype L_T constrains the pixel-level interclass relationship

$$D_{(i,j)} = \|X_T^{(i,j)} - L_T\left[Y_{\text{pseudo}}^{(i,j)} == c\right]\|_2 \tag{16}$$

where $D_{(i,j)}$ is defined as the feature distance in (i, j), $\|\cdot\|_2$ and $[\cdot]$ repersent l_2 norm and Iverson bracket, respectively.

Moreover, the interclass correlation matrix is derived from L_T and is employed to characterize the cross-domain interclass semantic interrelation

$$M_T = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1C} \\ r_{21} & r_{22} & \cdots & r_{2C} \\ \vdots & \vdots & \ddots & \vdots \\ r_{C1} & r_{C2} & \cdots & r_{\text{CC}} \end{bmatrix} \tag{17}$$

where r_{ck} is formulated as the Pearson correlation coefficient

$$r_{\text{ck}} = \frac{\mathbf{p}_c^\top \mathbf{p}_k}{\|\mathbf{p}_c\| \cdot \|\mathbf{p}_k\|}. \tag{18}$$

$D_{(i,j)}$ and M_T jointly conduct regularization guidance on the uncertain regions to obtain more confident and accurate pseudolabels. Their procedure of calibrating the pseudolabel is defined as

$$\hat{Y}_{\text{pseudo}}^{(i,j)} = \left\{ \underset{c\in 0,1,\ldots,C-1}{\arg\max}\, (D^{(i,j,c)} \cdot M_T^c \cdot Y_{\text{pseudo}}^{(i,j,c)}), \text{top} = K) \right\} \tag{19}$$

where K represents the number of classes which be compared, and K is set to 3 in this article. In contrast, the class with a higher weighted score is considered to be the correct classification result for the pixel.

E. Loss Function

The proposed HPL framework eliminates convoluted loss function engineering prevalent in existing approaches, achieving SOTA computational efficiency in both training convergence and inference execution. In brief, it is a straightforward end-to-end network in our proposed HPL

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}}^{\mathcal{D}_s} + \mathcal{L}_{\text{seg}}^{\mathcal{D}_t} + \lambda \mathcal{L}_{\text{domain}}. \tag{20}$$

IV. Experiments

This section focuses on the experimental setup, results, and discussion. It is divided into five sections: dataset, implementation details, ablation experiments, hyperparameter analysis, and comparisons with SOTA methods.

A. Datasets

Three datasets are employed to validate the proposed method under different sensors, landcover, and land form, including two public datasets *ISPRS Potsdam* and *LoveDA*, and a private dataset *Arcmap SX48*.

1) ISPRS Potsdam: Potsdam dataset [60] contains 38 RSIs with 6000×6000 pixels and 5 cm resolution of urban areas in the Potsdam region of Germany, accompanied by containing six classes labeled imagery. The Potsdam region is characterized by large building clusters, narrow streets, and dense settlement structures, with specific features labeled Impervious surfaces, building, low vegetation, tree, car, and clutter. In the UDA task under this dataset, the performance of the algorithms on different sensing is verified using **RGB** and **IRRG** cross-domain images of the same coverage area.

2) LoveDA: LoveDA dataset [61], provided by Wuhan University, contains 5987 RGB RSIs with 1024×1024 resolution of 0.3 m for **Urban** and **Rural** scenes in Nanjing, Changzhou, and Wuhan, China. In cross-domain scenes, both urban and rural, seven categories are included: background, building, road, water, barren, forest, and agriculture.

3) Arcmap SX48: The self-building dataset are all from the Arcmap platform [62], and the typical scenes of the central **Basin** and southern **Hills** regions of Shaanxi are selected as shown in Fig. 6, which contain eight RSIs with a spatial resolution of 4.8 m around shape of $12\,000 \times 12\,000$ pixels. They are divided into 6234 patches with 512×512 resolution. In this dataset, there are 12 categories of typical land-cover classes that are annotated, including cloud, farmland, grass, tree, village, urban, school, factory, greenhouse, ravine, road, and water.

B. Implementation Details

All methods are implemented with the DeepLabv3+ [63] model with ResNet-101 [64] backbone and initialized by a pretrained model with ImageNet-1k [65]. In the proposed method, the SGD optimizer with momentum of 0.9 and weight decay of $1e^{-4}$ is employed, and the learning rate is set to $5e^{-3}$ with poly of 0.9. In the dataset construction, each dataset is nonoverlappingly cropped into patches of size 512×512 pixels, and 80%, 10%, and 10% are randomly selected as the training, validation, and test set, respectively. In addition, a data enhancement strategy of randomly flipping 90°, 180°, and 270° is applied during the training process. The model is trained with a batch size of 8, and the number of training epochs on each dataset is set to 120 on the PyTorch framework with a single NVIDIA RTX 3090 GPU.

C. Ablation Studies

1) Discussion on Different Designed Modules: The effectiveness of the different modules on the network is analyzed with focus on three datasets as illustrated in Table I. The ablation experiment consists of the following four parts.

1) *SDPI:* SDPI attained an improvement ranging from 9.05% to 13.27% in mIoU on three selected datasets. The generic prototypes and the specific prototypes are uncoupled to facilitate intensive interaction, which elevates the network's perception capacity for spatial structures and semantic features.
2) *SDPI+DPA:* The additionally incorporated DPA promotes profound cross-domain alignment by prototype-interaction-based features. It attains an improvement

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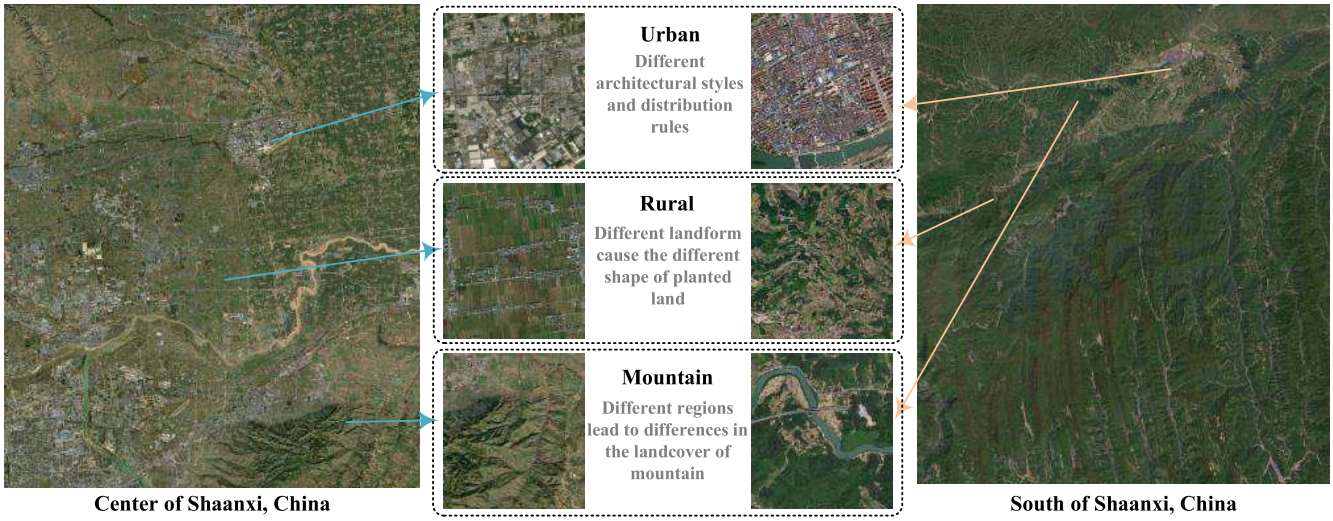


Fig. 6. Illustration of a pair of samples from the central and southern regions of Shaanxi Province, based on the Arcmap SX48 dataset with a spatial resolution of 4.8 m. Significant geographic differences between the Basin and the Hills are observed, especially in terms of woodland density, water diversity, village distribution, and urban structure. Moreover, differences in cultivated land shapes also contribute to cross-domain subtle variations.

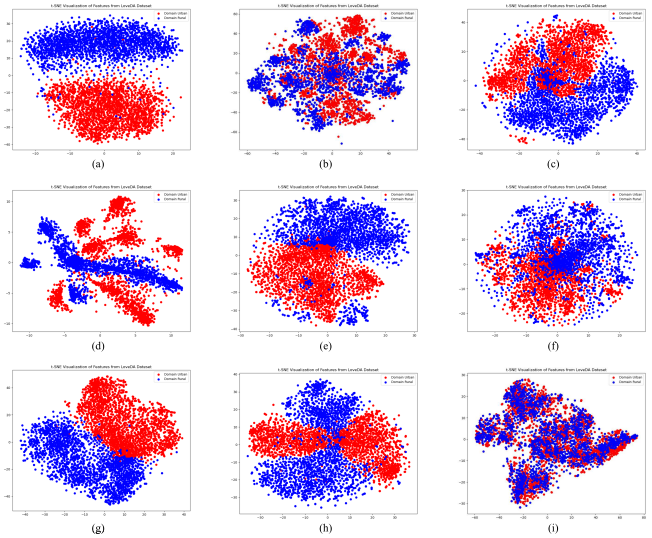


Fig. 7. t-SNE visualization results on cross-domain alignment of the last layer in the feature extraction module. (a) FADA. (b) DCA. (c) ProCA. (d) CLA-UDA. (e) MBATA. (f) MemoryAdaptNet. (g) MIC. (h) UemDA. (i) HPL.

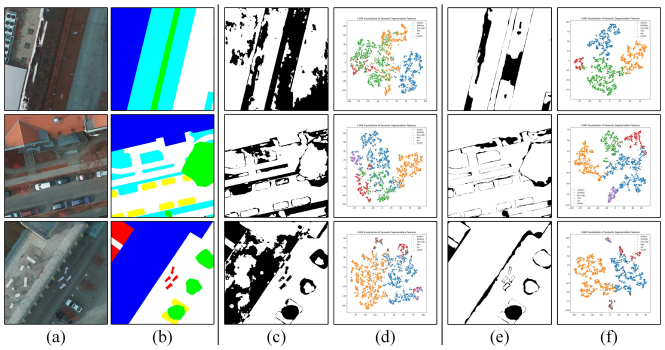


Fig. 8. Visualization results on classwise domain alignment of the proposed HPL. (a) RSIs. (b) GT. (c) Error map of baseline. (d) t-SNE visualization of baseline. (e) Error map of HPL. (f) t-SNE visualization of HPL. (c) and (e) Black area indicates the misclassification region.

varying between 9.05% and 16.29% in mIoU and between 8.61% and 13.47% in m$F1$. Moreover, to confirm the advantage of DPPI in aligning cross-domain differences, Fig. 7 illustrates the interdomain distributions of different UDA methods at the last layer of the feature extraction module. In this collection, the extracted HPL exhibits the best feature alignment aptitude.

3) *DCPE+AUC:* Given that DCPE and AUC are intensely coupled, it is inappropriate to evaluate the performance of DCPE in isolation. However, it is important to note that, in the absence of AUC, a fixed threshold of 0.7 is applied to enable the standalone DCPE to generate pseudolabels. Under the combined effect of DCPE and AUC, the mIoU of the ISPRS Potsdam, LoveDA, and Arcmap SX48 datasets has been increased by 14.53%, 14.78%, and 8.22%, respectively. In addition, t-SNE visualization result verifies that the proposed HPL has a significant effect on fine-grainedclass-level alignment in Fig. 8.

4) *SDPI+DPA+DCPE+AUC:* The proposed HPL, which consists of these four modules, comprehensively considers the hierarchical domain differences from both the domainwise and the classwise learning. The ablation experiments depicted in Fig. 7 illustrate the efficiency of HPL both within and between modules across the three datasets. The m$F1$ has been escalated by up to 19.27% on the LoveDA dataset, and the mIoU has been elevated by up to 22.73% on the ISPRS Potsdam dataset.

2) Discussion on Interaction in Different Stages: To compare the influence of features at different stages on domainwise feature alignment, in the backbone network, $S1$ and $S2$ are defined as shallow stages, while $S3$ and $S4$ are designated as deep stages. Table II demonstrates that while employing

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TABLE I
ABLATION EXPERIMENTS ON DIFFERENT MODULES UNDER ALL DATASETS

Task	DPPI		CDPC		mF1	mIoU
	SDPI	DPA	DCPE	AUC		
RGB ↓ IRRG	SO (Baseline)				59.6 (-)	43.2 (-)
	✓				65.71 (+6.11)	56.47 (+13.27)
	✓	✓			68.21 (+8.61)	59.49 (+16.29)
			✓		66.47 (+6.87)	52.3 (+9.10)
			✓	✓	70.17 (+10.57)	57.73 (+14.53)
	✓	✓	✓		76.42 (+16.82)	62.16 (+18.96)
	✓	✓	✓	✓	**78.27 (+18.67)**	**65.93 (+22.73)**
Urban ↓ Rural	SO (Baseline)				34.35 (-)	25.89 (-)
	✓				46.2 (+11.85)	36.27 (+10.38)
	✓	✓			47.82 (+13.47)	39.33 (+13.44)
			✓		42.33 (+7.98)	36.05 (+10.16)
			✓	✓	46.29 (+11.94)	40.67 (+14.78)
	✓	✓	✓		49.53 (+15.18)	42.78 (+16.89)
	✓	✓	✓	✓	**53.62 (+19.27)**	**45.77 (+19.88)**
Basin ↓ Hills	SO (Baseline)				50.56 (-)	32.93 (-)
	✓				56.44 (+5.88)	39.42 (+6.49)
	✓	✓			59.42 (+8.86)	41.98 (+9.05)
			✓		55.0 (+4.44)	38.7 (+5.77)
			✓	✓	58.71 (+8.15)	41.15 (+8.22)
	✓	✓	✓		62.17 (+11.61)	44.69 (+11.76)
	✓	✓	✓	✓	**64.64 (+14.08)**	**46.98 (+14.05)**

TABLE II
EXPERIMENTS ON INTERACTION IN DIFFERENT BACKBONE STAGES. THE BEST RESULTS ARE BOLD

Shallow stage		Deep stage		Task		
S1	S2	S3	S4	RGB→IRRG	Urban→Rural	Basin→Hills
✓		✓		64.89	45.23	45.83
✓			✓	**65.93**	**45.77**	**46.98**
	✓	✓		64.32	44.18	45.39
	✓		✓	65.66	45.42	46.76

$S3$ as the deep feature struggles to capture complete semantic representations, leading to limited effectiveness, using $S2$ as the shallow feature is insufficient to adequately model spatial interrelationships. This further demonstrates that deploying $S1$ and $S4$ as shallow and deep features, respectively, for interaction and cross-domain alignment achieves optimal outcomes on the three datasets.

D. Hyperparameters Analysis

1) σ Discussion: To validate AUC effectiveness in model optimization, experiments on the Arcmap SX48 dataset compared fixed thresholds (0.25, 0.5, 0.75, and 1) with the proposed dynamic thresholding strategy. Results in Table III show that the 0.25 threshold severely degraded performance due to ineffective pseudolabels and overfitting, while 0.5 achieved the closest results to HPL, suggesting its suitability for this dataset. However, HPL outperformed fixed thresholds with the highest mIoU and optimal results in 8 of 12 classes by dynamically adjusting thresholds during training. Further experiments across three datasets confirmed that optimal thresholds vary significantly by dataset, as shown in Table IV, requiring laborious tuning for fixed thresholds, which contradicts transfer learning principles. The findings highlight the critical impact of threshold magnitude on model performance and demonstrate the superiority of adaptive threshold adjustment over static approaches.

2) λ Discussion: As shown in Table V, we performed a sensitivity experiment on λ with key values. The results indicated a notable performance decline as λ decreased from 1 to 0.1. Conversely, within the range of 0.05 to 0.001, the effectiveness remained stable and desirable. Among the five fine-grained values (0.05, 0.02, 0.01, 0.005, and 0.001), setting λ to 0.01 exhibited superiority across utilized three datasets utilized.

E. Comparisons With SOTA Methods

To highlight the advantages of the proposed method, we make comparisons with the existing SOTA methods, which consist of AdaptSegNet [34], FADA [21], DCA [66], ProCA [57], CLA-UDA [67], MBATA [14], MemoryAdaptNet [15], MIC [40], and UemDA [51]. AdaptSegNet is the classical domain-adaptive semantic segmentation method using adversarial learning. DCA, CLA-UDA, and MBATA are SOTA methods with improved adversarial learning, while FADA, ProCA, and MemoryAdaptNet are SOTA methods introducing prototype learning for UDA semantic segmentation. MIC and UemDA exhibited the recent RSISS research in UDA of self-training.

1) ISPRS Potsdam: Table VI illustrates the quantitative results of the different methods on the ISPRS Potsdam dataset. In the absence of DA methods, the efficiency of SO in each class is not satisfactory, and the mIoU is only 43.20%. The proposed HPL attained the optimal results in 5 out of six classes. Compared to the second-best results of each class, the IoU for *surface*, *building*, *tree*, *car*, and *clutter* increased by 0.74%, 0.16%, 0.78%, 3.76%, and 0.19%, respectively. It also achieved the best mIoU with 65.93%, which is 22.73% and 0.9% higher than SO and the second-best MemoryAdaptNet, respectively. They demonstrate that domainwise and classwise prototype learning play a significant role in alleviating domain divergence.

Fig. 9 demonstrates the ISPRS Potsdam visualization outcomes. The IRRG imagery exhibits a predominant red coloration due to spectral variations between the B (blue) and IR (infrared) bands, differing distinctly from standard RGB images. This chromatic discrepancy reflects sensor-induced domain shifts between multispectral and conventional imaging systems. As shown in Fig. 9(c), RSISS without UD strategy gives rise to grave obscuring of class boundary and acute ambiguity between different classes. In the SO method, the boundary of *buildings* is not accurately depicted in the first and third rows. In the upper right corner of the second row, there is confusion between *trees*, *low vegs*, and *buildings*. The fifth row shows clear misclassifications between *clutter*, *buildings*, and *low vegs*, and it is also unable to accurately distinguish between *clutter* and *Impervious*. As shown in Fig. 9(k) and (l), in comparison with remaining SOTA methods, the proposed HPL exhibits a substantial enhancement in edge definition and additionally lessens between-class confusion. In addition to addressing the issues of the previous methods, the boundary between *Impervious* and *low vegs* in the second column is accurately outlined, and the sixth column shows the ability to

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TABLE III

ABLATION EXPERIMENTS COMPARE THE PROPOSED AUC WITH DIFFERENT σ UNDER ARCMAP SX48 DATASET. THE BEST RESULTS ARE BOLD

Task	α	Class												mIoU (%)
		cloud	farmland	grass	tree	village	urban	school	factory	greenhouse	ravine	road	water	
Basin ↓ Hills	0.25	48.78	40.59	18.04	62.58	41.34	27.29	32.29	39.94	42.18	18.33	15.13	38.79	35.44
	0.50	60.33	47.72	19.65	**82.37**	52.92	**48.7**	58.45	**42.32**	52.51	29.18	16.42	46.01	46.38
	0.75	63.81	46.11	22.72	76.54	46.92	39.06	42.63	41.99	51.1	24.27	19.43	43.85	43.20
	1.00	62.35	43.42	**24.01**	74.21	46.78	41.31	41.92	41.42	48.25	24.37	18.7	42.89	42.47
	Ours (APU)	**64.79**	**49.56**	23.63	77.36	**53.09**	42.44	**59.02**	41.46	**55.67**	**29.43**	**20.08**	**47.25**	**46.98**

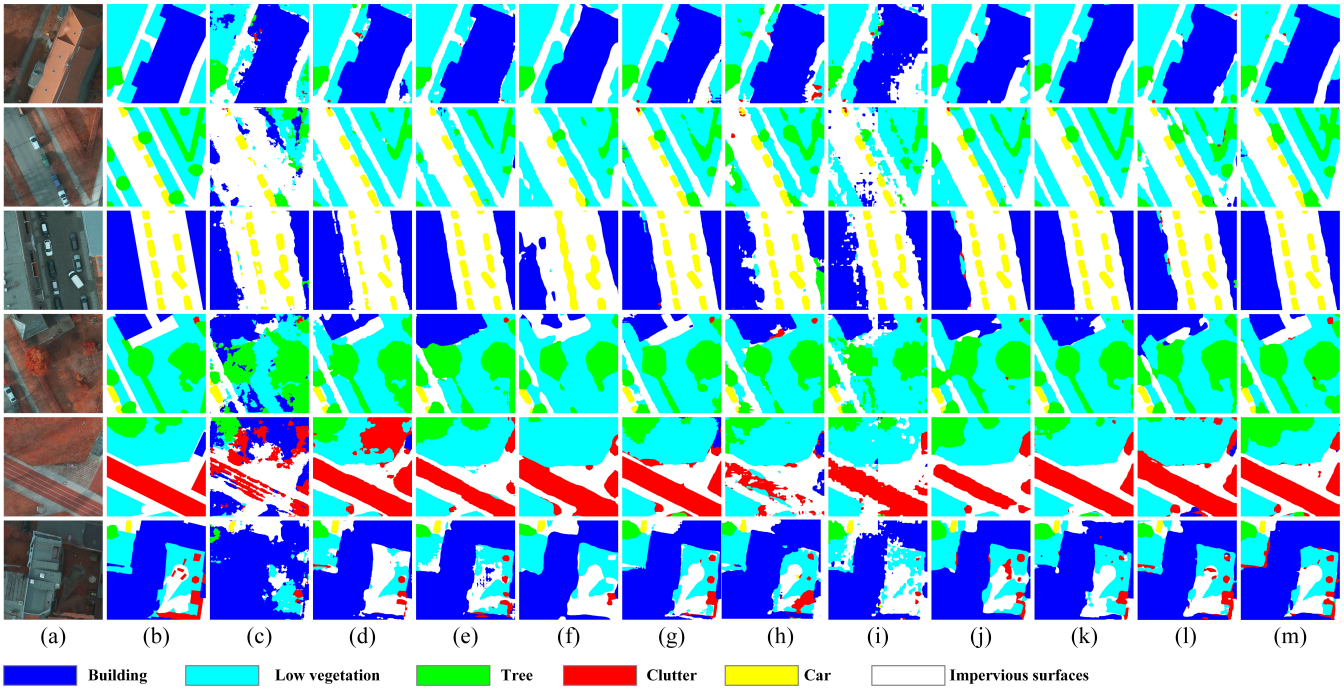


Fig. 9. Visualization results of the ISPRS Potsdam dataset. (a) Original RSIs. (b) GT. (c) Source only. (d) AdaptSegNet. (e) FADA. (f) DCA. (g) ProCA. (h) CLA-UDA. (i) MBATA. (j) MemoryAdaptNet. (k) MIC. (l) UemDA. (m) Proposed HPL.

TABLE IV

ABLATION EXPERIMENTS OF OPTIMAL FIXED AND AUC RESULTS ON DIFFERENT DATASETS. THE BEST RESULTS ARE BOLD

Dataset	σ Type	mIoU
Potsdam	Fixed 0.75	65.25
	AUC	**65.93**
LoveDA	Fixed 0.5	43.38
	AUC	**45.77**
Arcmap SX48	Fixed 0.5	46.38
	AUC	**46.98**

partition a larger portion of *clutter*. This proves the advantage of the pseudolabel updating strategy for uncertain pixels.

2) LoveDA: Table VII exhibits the results of the evaluation metrics for the LovDA dataset. The proposed HPL method, which employs prototype learning as an essential technique, makes use of domain-level and class-level prototypes to enhance the model effectively, reaching an mIoU of 45.77%. It represents a 19.88% improvement over the SO method. In terms of specific classes, the segmentation accuracy of the algorithms for each class should be observed from the perspective of stability, although DCA, CLA-UDA, and MBATA achieved the best results in *agriculture*, *water*, and *background*, respectively. Although DCA, CLA-UDA, and MBATA attain the optimal IoU for *agriculture*, *water*, and *background*, respectively. There is a severe disparity in classification accuracy for *barren* and *road*. The proposed HPL obtains the optimal IoU for the categories of *building*, *road*, *barren*, and *forest*. There are four classes (*building*, *road*, *barren*, and *forest*), but they also have relatively satisfactory performance in the rest of the classes, with the lowest IoU of 34.69% for *background*. Therefore, the superiority and stability of the algorithm demonstrate that the proposed HPL accomplishes the best performance on this dataset.

Fig. 10 displays the segmentation visualization results for all compared methods. AdaptSegNet exhibits excessive noise in local areas during the segmentation process. The results from CLA-UDA and MBATA present significant edge roughness, especially for the *building*, *barren*, and *agriculture* classes. The visualization results of DCA are generally excellent, but

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TABLE V
EXPERIMENTS OF DIFFERENT λ ON DIFFERENT DATASETS. THE BEST RESULTS ARE BOLD

Task	λ							
	1	0.5	0.1	0.05	0.02	0.01	0.005	0.001
RGB→IRRG	59.70	62.39	62.74	65.17	65.89	**65.93**	65.09	64.84
Urban→Rural	35.23	39.71	42.14	44.61	45.30	**45.77**	44.82	43.73
Basin→Hills	37.89	40.42	41.37	45.65	46.92	**46.98**	45.45	45.01

TABLE VI
EXPERIMENTAL METRICS FOR THE ISPRS POTSDAM DATASET. THE BEST RESULTS ARE BOLD

Task	**method**	**Class**						**mIoU**
		surface	**building**	**low vegs**	**tree**	**car**	**clutter**	
RGB ↓ IRRG	SO [63]	56.73	58.29	26.20	48.20	56.32	13.43	43.20
	AdaptSegNet [34]	65.63	79.82	46.49	59.78	72.39	25.03	58.19
	FADA [21]	69.11	81.91	56.70	62.79	75.21	25.83	61.93
	DCA [66]	68.16	80.46	62.92	61.81	73.03	23.03	61.57
	ProCA [57]	72.23	83.88	57.21	63.26	76.51	30.14	63.87
	CLA-UDA [67]	64.13	78.25	58.00	58.33	71.60	19.75	58.34
	MBATA [14]	63.01	76.44	50.63	57.22	72.66	28.78	58.12
	MemoryAdaptNet [15]	71.91	86.49	62.50	63.85	76.35	29.05	65.03
	MIC [40]	73.94	84.42	60.53	62.14	76.47	31.09	64.77
	UemDA [51]	70.32	81.45	**62.96**	59.79	75.20	28.98	63.12
	Ours	**74.68**	**86.65**	58.09	**64.63**	**80.27**	**31.28**	**65.93**

TABLE VII
EXPERIMENTAL METRICS FOR THE LOVEDA DATASET. THE BEST RESULTS ARE BOLD

Task	**method**	**Class**							**mIoU**
		background	**building**	**road**	**water**	**barren**	**forest**	**agriculture**	
Urban ↓ Rural	SO [63]	46.15	30.80	23.29	35.00	5.29	7.93	32.77	25.89
	AdaptSegNet [34]	39.80	36.05	31.02	34.61	16.81	20.16	15.48	27.70
	FADA [21]	48.25	38.35	28.04	46.94	16.51	20.98	40.43	34.21
	DCA [66]	48.35	31.27	38.43	48.51	14.46	22.67	**65.81**	38.50
	ProCA [57]	49.63	42.15	33.12	50.81	24.15	22.39	36.18	36.92
	CLA-UDA [67]	38.89	40.69	25.00	**55.63**	13.44	28.21	52.03	36.27
	MBATA [14]	**52.80**	39.44	31.85	34.73	18.64	21.29	19.73	31.21
	MemoryAdaptNet [15]	45.08	42.34	40.08	43.25	23.43	26.52	30.46	35.88
	MIC [40]	48.34	37.45	29.91	52.07	17.34	33.95	33.84	36.13
	UemDA [51]	49.03	44.43	37.39	49.22	14.95	27.14	47.28	38.49
	Ours	34.69	**56.56**	**50.36**	52.62	**40.39**	**42.20**	43.57	**45.77**

it is evident that there exists significant over-segmentation of boundaries for *building* and *water*. FADA, ProCA, and MemoryAdaptNet demonstrate satisfactory performance in the segmentation of local *building* regions, although they all exhibit erroneous classification of *agriculture* as *building*. The proposed HPL is competent to precisely separate *road* and the tiny regions of *building* and *water*. Furthermore, HPL exhibits the slightest rate of misclassification compared to other methods.

3) Arcmap SX48: Fig. 6 highlights geographic contrasts between Basin and Hills, showing differences in forest density, water diversity, settlement distribution, urban layouts, and farmland types. These nuances demand finer class partitioning in complex cross-domain scenarios, intensifying requirements for model adaptability and robustness.

Table VIII shows the evaluation metrics of different methods for the semantic segmentation task on the SX48 dataset. Compared with other methods, the proposed HPL has achieved the optimal values among all methods with 46.98% and 46.63% for mIoU and mIoU*, respectively. In addition, *farmland*, *village*, *factory*, *greenhouse*, *ravine* and *water* exhibit the best IoU, which are 4.21%, 11.99%, 22.64%, 20.56%, and 0.96% higher than those of the baseline SO, respectively. HPL is demonstrated to be the most reliable in both full and local cross-domain migration capability.

Fig. 11 exhibits the visualization of the segmentation results of different algorithms for typical regions on the Arcmap SX48 dataset. In order to visualize and compare the effectiveness of the algorithms in different regions, the results can be divided into three groups from top to bottom according to *Mountainous region*, *Urban region*, and *Rural region*.

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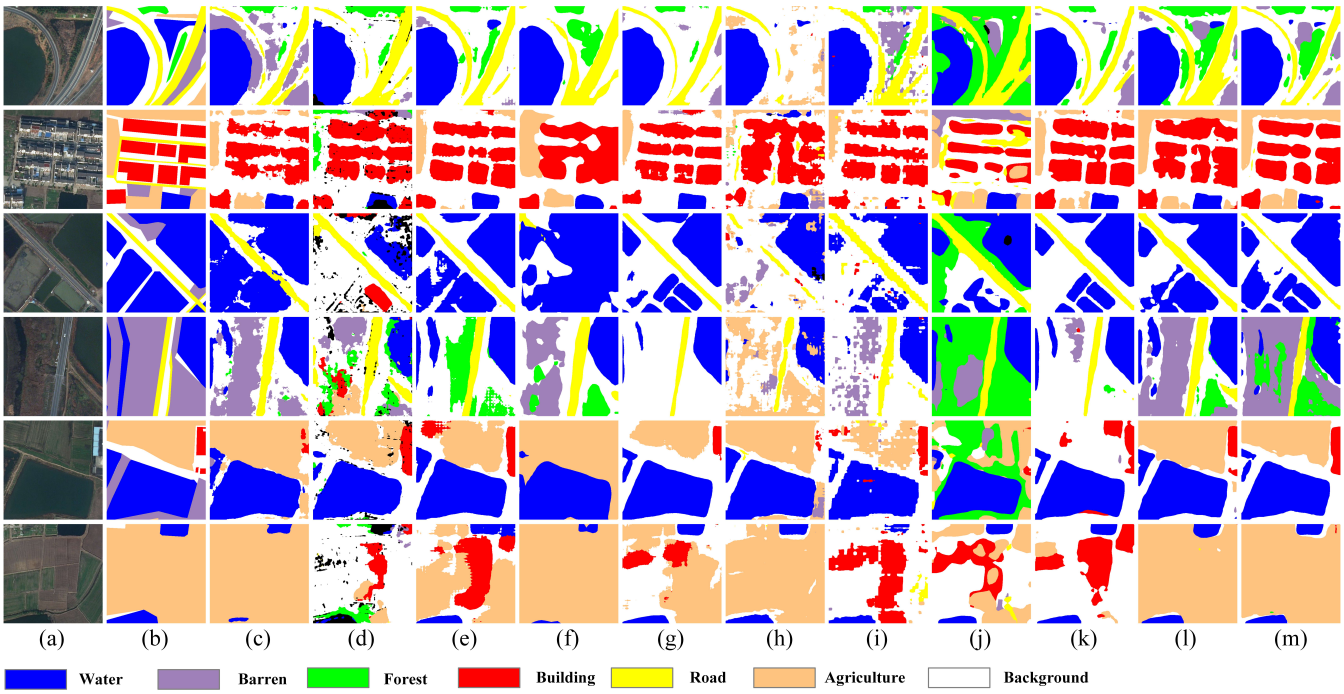


Fig. 10. Visualization results of the LoveDA dataset. (a) Original RSIs. (b) GT. (c) Source only. (d) AdaptSegNet. (e) FADA. (f) DCA. (g) ProCA. (h) CLA-UDA. (i) MBATA. (j) MemoryAdaptNet. (k) MIC. (l) UemDA. (m) Proposed HPL.

TABLE VIII

EXPERIMENTAL METRICS FOR THE ARCMAP SX48 DATASET. THE BEST RESULTS ARE BOLD

Task	method	Class												mIoU	mIoU*
		cloud	farmland	grass	tree	village	urban	school	factory	greenhouse	ravine	road	water		
Basin ↓ Hills	SO [63]	33.16	45.35	17.11	76.97	48.14	33.92	21.11	29.47	33.03	8.87	14.60	33.47	32.93	36.91
	AdaptSegNet [34]	30.98	42.44	17.76	55.14	49.35	23.58	41.14	37.32	52.63	9.97	19.54	39.52	34.95	35.07
	FADA [21]	47.86	46.99	16.31	75.40	50.14	35.37	30.30	39.62	28.65	14.18	15.53	36.60	36.41	40.42
	DCA [66]	9.01	42.37	16.46	81.10	43.51	50.88	-	38.19	36.45	-	9.15	28.63	35.58	35.48
	ProCA [57]	63.14	46.91	19.67	73.29	47.36	45.72	43.19	38.87	26.97	12.41	17.40	44.60	39.96	44.11
	CLA-UDA [67]	**65.90**	47.59	18.84	75.48	40.27	32.81	-	26.34	-	7.75	11.04	28.88	35.49	38.57
	MBATA [14]	45.71	45.62	**25.26**	72.33	45.41	42.07	28.12	24.94	17.12	29.19	7.73	26.37	34.16	37.27
	MemoryAdaptNet [15]	62.36	34.33	23.07	**82.43**	49.59	39.45	60.03	40.79	51.35	21.48	12.66	37.55	42.92	42.47
	MIC [40]	44.20	43.01	7.43	63.16	48.45	46.30	**73.40**	35.55	38.69	11.39	**21.70**	46.29	39.96	39.57
	UemDA [51]	39.13	41.73	18.40	79.45	45.31	**50.93**	57.98	40.53	39.04	18.72	18.25	40.16	40.80	41.54
	Ours	64.79	**49.56**	23.63	77.36	**53.09**	42.44	59.02	**41.46**	**55.67**	**29.43**	20.08	**47.25**	**46.98**	**46.63**

TABLE IX

MODEL COMPLEXITY OF THE PROPOSED METHOD AND THE SOTA METHODS. THE BEST RESULTS ARE BOLD

Methods	AdaptSegNet	FADA	DCA	ProCA	CLA_UDA	MBATA	MemoryAdaptNet	MIC	UemDA	Ours
Parms(M)	67.399	67.573	88.654	**58.627**	**58.627**	143.905	61.397	**58.627**	60.03	58.759
Flops(G)	98.888	149.210	90.980	**50.293**	249.259	173.204	76.138	100.600	50.372	158.647

In the results of *Mountainous region*, the majority of methods are competent to identify the overall class outlines of RSIs. The proposed HPL outperforms existing methods in preserving holistic scene completeness and accurately segmenting local objects. In the results of *Urban region*, due to the diverse differences between urban and mountainous regions, two typical cases are presented to illustrate the comparative visual: 1) the intricate distribution of roads and 2) different building distribution rule. Benefiting from the superior quality of the pseudolabel, it is obvious that the proposed HPL substantially surpasses the other comparison methods in terms of the integrity of the edges of each class. In the results of *Rural region*, HPL outperforms the other comparative methods in the segmentation of *factory* edge, *road* connectivity, and *village* boundary.

4) Model Complexity Analysis: In order to thoroughly evaluate the value of the model, we adopt the number of parameters (Params) to measure the model complexity and the giga floating-point operations per second (GFLOPs) to measure the theoretical computational overhead. We have

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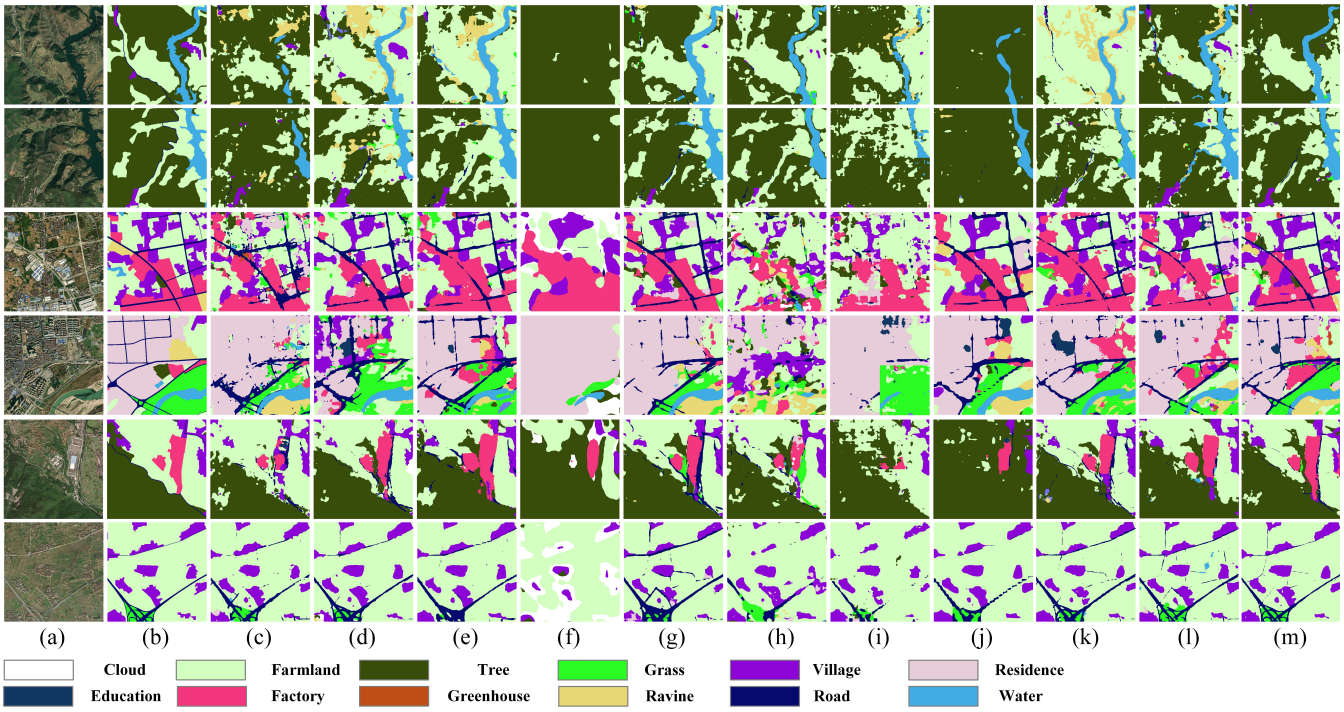


Fig. 11. Visualization results of the Arcmap SX48 dataset. (a) Original RSIs. (b) GT. (c) Source only. (d) AdaptSegNet. (e) FADA. (f) DCA. (g) ProCA. (h) CLA-UDA. (i) MBATA. (j) MemoryAdaptNet. (k) MIC. (l) UemDA. (m) Proposed HPL.

TABLE X
BIDIRECTIONAL DOMAIN ADAPTATION EXPERIMENTS WITH THREE SOTA METHODS. THE BEST RESULTS ARE BOLD

Method	IRRG→RGB		Rural→Urban		Hills→Basin	
	mF1	mIoU	mF1	mIoU	mF1	mIoU
MemoryAdaptNet [15]	80.64	67.23	49.89	45.98	67.33	47.45
MIC [40]	80.38	67.34	52.58	46.71	66.27	45.83
UemDA [51]	79.40	66.58	51.81	45.32	68.02	47.81
Ours	**81.55**	**68.72**	**56.67**	**51.13**	**68.09**	**48.25**

compared the model complexity and time complexity of the proposed method and the other models as shown in Table IX. The GFLOPs are calculated under the same conditions, and the input images are all set to $1 \times 3 \times 512 \times 512$. While our method demonstrates competitive performance in other quantitative metrics, its parameter count and computational cost do not rank top in this table. These results objectively reflect the ongoing tradeoff space between performance and complexity, highlighting room for further exploration to refine efficiency without compromising efficacy.

5) Bidirectional Experiments: To comprehensively assess the robustness of the proposed HPL across diverse datasets, three SOTA methods [15], [40], [51] were employed for comparison in the bidirectional domain adaptation experiments. As shown in Table X, the quantitative metrics of all algorithms demonstrate improvements across the three datasets. The ISPRS Potsdam dataset benefits from the adaptability of ImageNet pretrained weights, while the LoveDA and Arcmap SX48 datasets leverage the inherent regularities of urban structures. Notably, compared to the SOTA methods, the proposed HPL achieves superior performance in both mF1 and mIoU metrics. This is particularly evident in the LoveDA dataset, where the HPL outperforms the second-best method by 4.09% in m$F1$ and 4.42% in mIoU.

V. CONCLUSION AND FUTURE WORK

In this article, we propose a novel hierarchical prototype network with uncertainty-aware adaptation for RSISS in UDA, which leverages DPPI and CDPC to bridge the cross-domain discrepancy. Inherent domain consistency and class invariance of the SD and TD are completely investigated, and the liability of domain deviation arising from uncertainty regions of the pseudolabel is effectively addressed. Extensive experiments and analysis on two selected public datasets and a private dataset have proved the effectiveness of the proposed HPL.

Though prototype learning enables the extraction of discriminative features from a single dataset, its generalization across different datasets remains limited. Future work will focus on integrating prototype representations into pretrained models for RSI analysis and exploiting geographic prior knowledge to facilitate prototype learning so as to enhance its generalization. On the other hand, multimodal information

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(such as visual-language) is given priority consideration to further augment cross-domain-adaptive capabilities.

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Jiawei Ning received the B.S. degree in computer science and technology from the Mingde College, Northwestern Polytechnical University, Xi'an, China, in 2014, and the master's degree in computer science and engineering from the School of Computer Science and Engineering, Xi'an University of Technology, Xi'an, in 2022. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Xidian University, Xi'an.

His research interests include transfer learning and remote sensing image semantic segmentation.



Zhongle Ren (Member, IEEE) received the B.S. degree in electronic and information engineering and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2013 and 2020, respectively.

He is currently a Lecturer at the School of Artificial Intelligence, Xidian University. His interests include large-scale remote sensing pretraining models, and synthetic aperture radar (SAR) images understanding and interpretation.



Biao Hou (Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University, where he is currently a Professor. His research interests include compressive sensing and synthetic aperture radar image interpretation.



Runnong Jiang received the B.S. degree in computer science and technology from the School of Computer Science and Technology, Anhui University, Hefei, China, in 2023. He is currently pursuing the M.S. degree in computer technology with Hangzhou Institute of Technology, Xidian University, Xi'an, China.

His research interests include optical remote sensing image pretraining and optical remote sensing image semantic segmentation.

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Shijie Liu received the B.S. degree in intelligent science and technology from the School of Artificial Intelligence, Xidian University, Xi'an, China, in 2022, where he is currently pursuing the Ph.D. degree in computer science and technology.

His research interests include the development of pretrained models for remote sensing applications and methods for imbalanced land-cover classification.



Weibin Li received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1998 and 2000, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2004.

He is currently the Director of the Artificial Intelligence Laboratory, Hangzhou Institute of Technology, Xidian University, where he is currently a Professor. His research interests include remote sensing image processing and high-precision positioning technology and applications.



Licheng Jiao (Life Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China and the International Research Center of Intelligent Perception and Computation. His research interests include intelligent information processing, image processing, machine learning, and pattern recognition.

Dr. Jiao is a member of the IEEE Xi'an Section Execution Committee, the President of the Computational Intelligence Chapter, the IEEE Xi'an Section, and the IET Xi'an Network, the Chairperson of the Awards and Recognition Committee, the Vice Board Chairperson of the Chinese Association of Artificial Intelligence, a Councilor of Chinese Institute of Electronics, a Committee Member of Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.

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